

TED (10) – 5007

Reg.No.....

(Revision -2010)

(Model Paper)

Signature.....

Fifth Semester Diploma Examination

Quantity Surveying & Construction Management

HIGH WAY ENGINEERING

(Maximum marks -100)

Time: 3 Hours

I Part – A (Answer in one or two sentences – each question carry 2 marks)

- a) What is the duration of road plans? Specify the period of current road plan.
- b) What is the purpose of providing camber to road surface?
- c) On conducting aggregate crushing value test on a sample weighing 2000gms, the weight of aggregates passing through 2.36mm sieve was obtained as 500gms. What is the crushing value of the sample?
- d) For what purpose sheep foot rollers are used.
- e) List any four mandatory signals.

II Part –B (Answer any five – each question carry 6 marks)

- a) Discuss the salient features of the current road plan?
- b) Discuss the suitability, advantages and limitations any three road pattern
- c) What are the elements in the geometric design of road?
- d) What is meant by breaking sight distance? What is its significance in fixing alignment
- e) What are the different machineries used in road construction. Mention their uses.
- f) Why signaling system is required in traffic system? What are the characteristics of an ideal signaling system?
- g) What are the significant road user characteristics?

Part – C (Answer any one full question from each module)

(Each subdivision carries 5 marks)

Module-I

- III**
- a) In what way the recommendations of Jayakar committee report was implimented?
 - b) What are the objects of reconnaissance survey?
 - c) What is the advantage of Inter Modal mix of traffic?

OR

- IV**
- a) What are the objects of final location survey?
 - b) What are the details collected during the economic studies?
 - c) What are the data collected during the Engineering Surveys?

Module-II

- V**
- a) How the sub grade is prepared for a new road?
 - b) Under what circumstance do we provide a horizontal curve, summit curve and a valley curve?
 - c) Differentiate between Right of way, Carriageway and Roadway

OR

- VI**
- a) What is meant by supper elevation? Why is it necessary to provide supper elevation?
 - b) Differentiate between simple curve, compound curve and transition curve.
 - c) Draw the cross section of a hill road partially in cutting and filling.

Module-III

- VII**
- a) How the abrasion resistance of aggregate is determined by Los Angeles abrasion Test
 - b) What are the ideal characteristics of soil preferred for sub grade?
 - c) Differentiate between tack coat and seal coat

OR

- VIII**
- a) Explain the construction steps of penetration Macadam.
 - b) What are the recommendations of Concrete Association of India with regard to concrete roads?
 - c) What is meant by soil stabilized roads? How bitumen stabilized roads are made

Module-IV

- IX**
- a) How the maintenance of bituminous roads are carried out during monsoon season?
 - b) What are the different types of parking arrangements provided? Explain the advantages and limitations of each type.
 - c) What are the peculiar features of Express Highways

OR

- X**
- a) Mention any three traffic control device and explain their function?
 - b) Draw the sketch of a road and mark the following a) Boundary limit, Centre line, No pass zone, Pedestrian Crossing
 - c) What is the basis for the classification of traffic signals?